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Smart EV Charging Station Queue and Slot Scheduling System

CSA0604 - DESIGN AND ANALYSIS OF ALGORITHMS

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Introduction

- The **Smart EV Charging Station Queue and Slot Scheduling System** is designed to manage the increasing demand for electric vehicle (EV) charging stations.
- As the number of EVs increases, charging stations can experience **long queues, limited charging slots, and increased waiting time**.
- Traditional **First-Come-First-Served (FCFS)** systems may not efficiently handle different charging requirements, battery levels, and urgent users.
- The proposed system allows EV users to **book charging slots in advance** and receive suitable charging schedules.

Problem Statement

- Existing charging stations experience long waiting times during peak hours.
- Manual slot allocation makes the charging process inefficient.
- Users cannot easily know the real-time availability of charging slots.
- Poor queue management results in wasted time and energy.
- An intelligent scheduling system is required to improve charging efficiency.





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Objectives

- To manage EV users, vehicles, charging stations, and slots efficiently.
- To provide secure registration and easy vehicle information management.
- To schedule charging slots and manage vehicle queues effectively.
- To monitor charging activities and generate useful system reports.

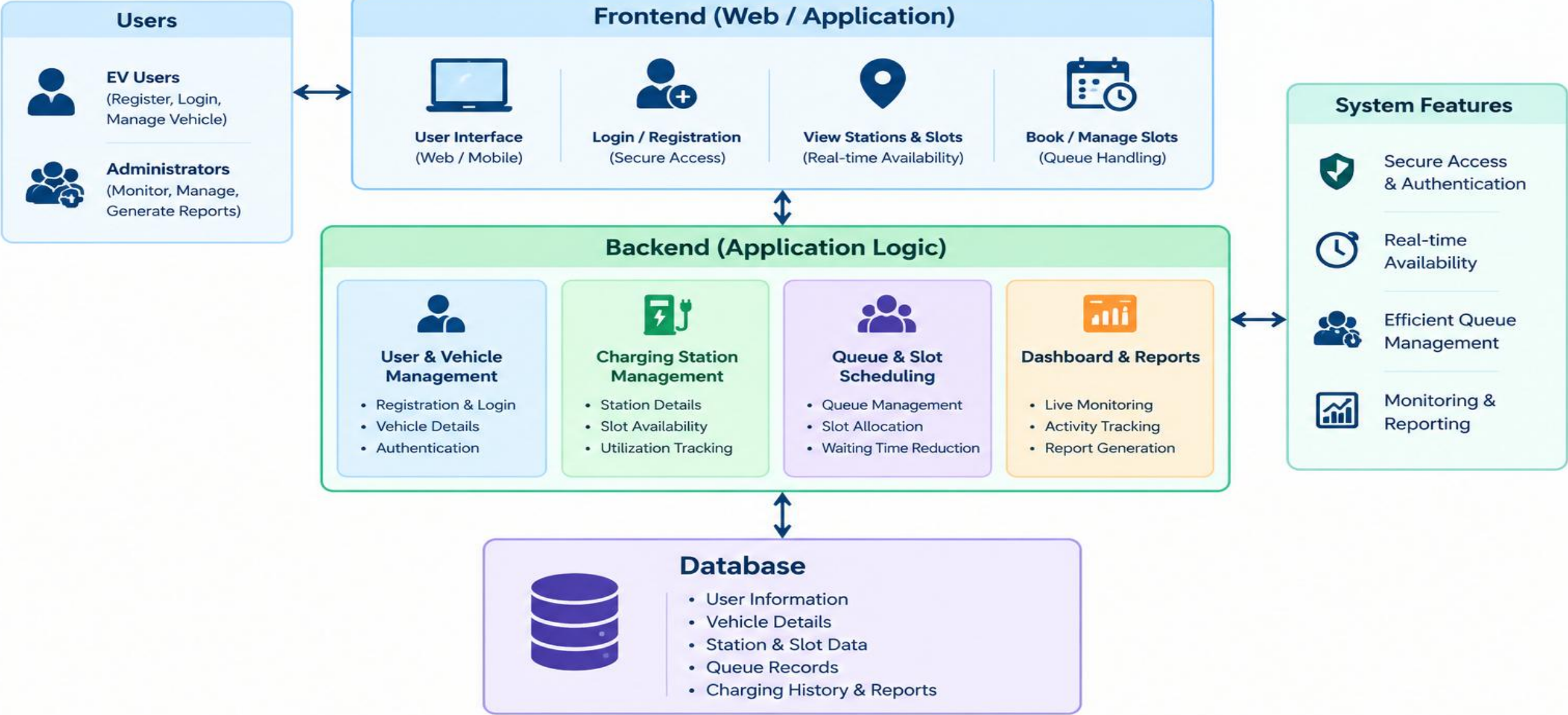


Literature Review

Author / Year	Research Focus	Approach	Key Finding	Gap / Limitation
Wang et al., 2025	EV charging queue management	Queuing model + bi-level scheduling	Improved charging demand fulfillment and reduced queuing time.	Focuses strongly on power constraints and optimization. (MDPI)
Varshney et al., 2025	EV charging station queues	Stochastic modeling + queueing analysis	Models waiting, abandonment, station utilization and throughput.	More analytical; does not provide a complete user-facing management system. (Nature)
Upadhyaya et al., 2026	Fast-charging scheduling	Online scheduling + optimization	Reported reduced waiting time under limited charger availability.	Mainly concentrates on scheduling optimization rather than integrated station management. (Taylor & Francis Online)
Liu & Zhu, 2026	EV charging allocation	Two-stage allocation + congestion-aware queuing	Uses station-level allocation to control congestion and improve charging assignment.	Primarily focuses on network-level allocation. (arXiv)
Kahlert et al., 2026	Charging congestion forecasting	Fluid-queue forecasting	Forecasting congestion can improve charging schedules and reduce waiting-related downtime.	Focuses on forecasting rather than complete queue, slot and dashboard management. (arXiv)

System Architecture

Smart EV Charging Station Queue and Slot Scheduling System





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Module-1: User Registration and Login Module

- Allows new users to register in the system.
- Stores user details such as name, email, phone number, and EV details.
- Provides secure login using username/email and password.
- Allows users to update their profile information.
- Maintains user account and vehicle information.
- Provides access to booking and charging services after login.


VoltQueue
Smart EV Charging · Queue & Slot Scheduling

[→ Sign in](#) [Create account](#)

Email
you@voltqueue.io

Password

[→ Sign in to VoltQueue](#)

 DEMO CREDENTIALS — ONE CLICK TO FILL

 Station Administrator	ADMIN
admin@voltqueue.io / admin123 Full network control · charger fleet · simulation	
 EV Driver	USER
user@voltqueue.io / user123 Book slots · track tickets · watch the live queue	

 ADMIN unlocks the network dashboard, charger fleet management and the simulation sandbox. Drivers can browse, book slots and track tickets.



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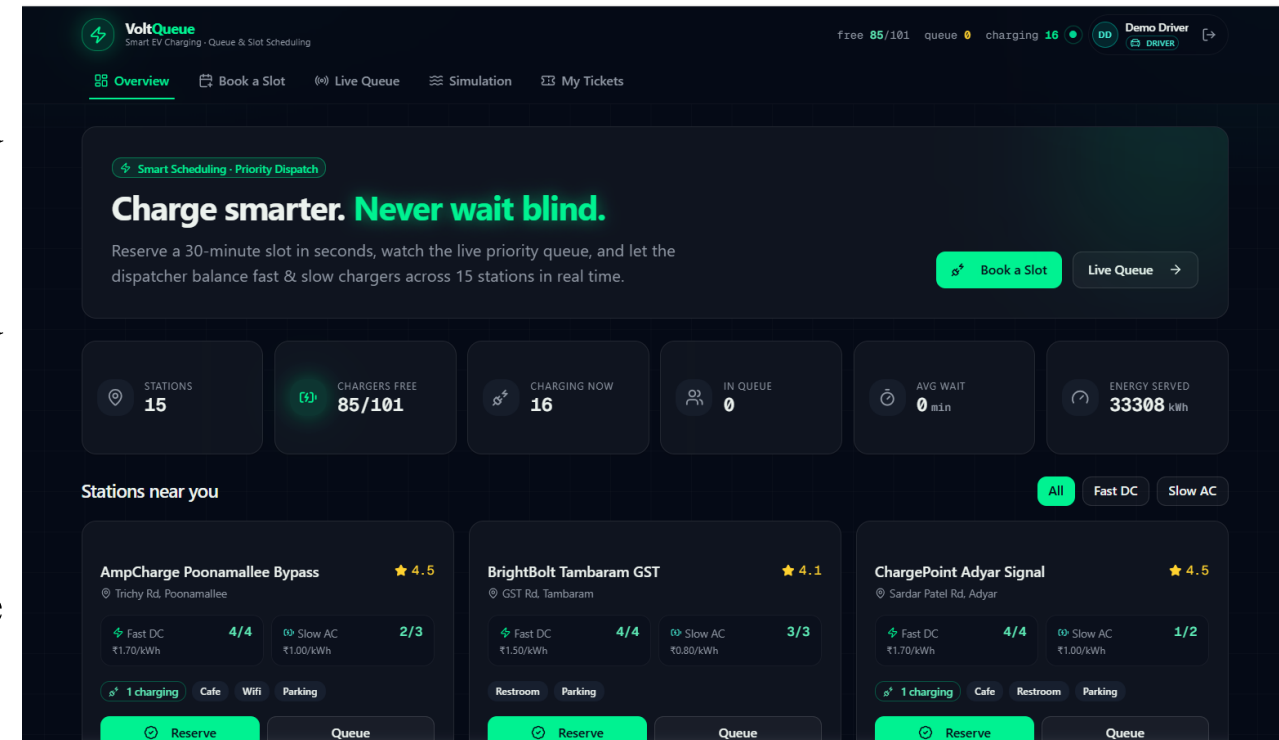


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Module 2: Charging Station and Slot Management Module

- Manages available EV charging stations.
- Displays the number of available and occupied charging slots.
- Stores charger details such as charger type and charging capacity.
- Updates slot status in real time.
- Allows administrators to add, update, or remove charging stations.
- Helps users view available charging slots.





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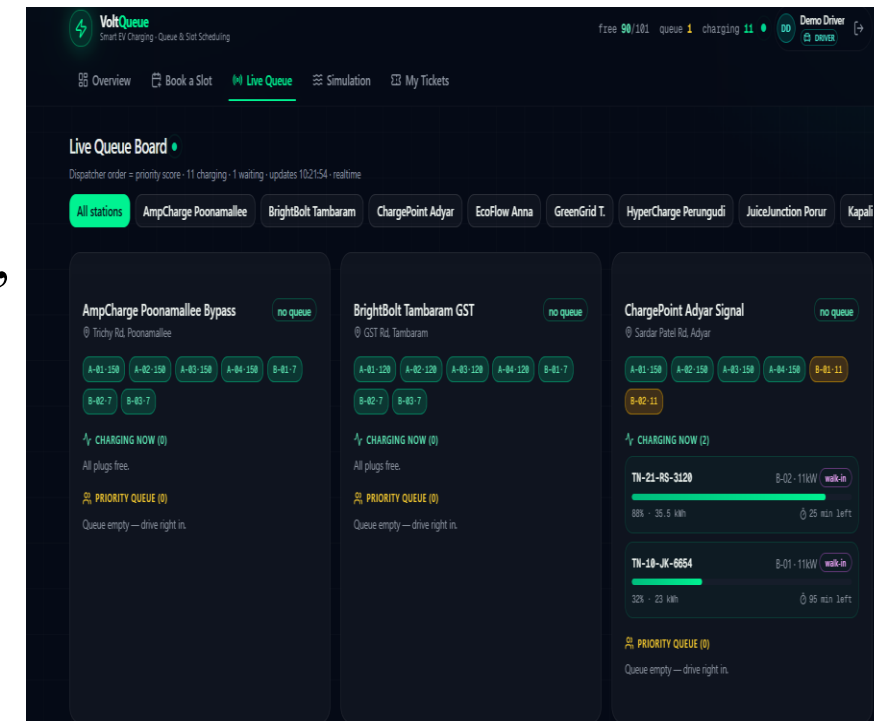


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Module 3: Charging Slot Allocation and Queue Management Module

- Allocates charging slots based on availability.
- Maintains a queue when all charging slots are occupied.
- Considers arrival time, battery level, charging duration, and priority.
- Reduces unnecessary waiting time for EV users.
- Automatically assigns the next available slot.
- Supports cancellation and rescheduling of bookings.





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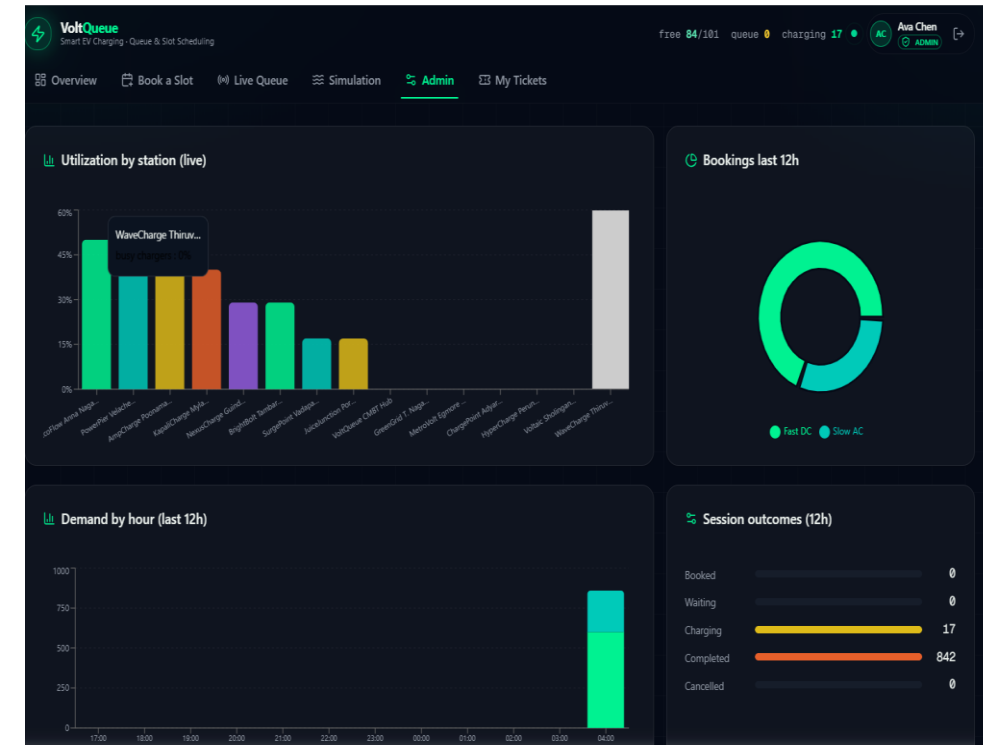


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Module-4: Charging Monitoring and Report Generation Module

- Monitors ongoing charging activities.
- Records charging start time and completion time.
- Tracks charging duration and energy consumption.
- Displays the current status of each charging session.
- Generates daily, weekly, and monthly reports.
- Provides information about station usage, waiting time, and completed charging sessions.





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Tools and Technologies

Component	Technology / Tool Used
AI-Assisted Development	Z.AI
Frontend Development	HTML5, CSS3
Client-Side Scripting	JavaScript
Backend / Server-Side	Web-based Server Environment
Data Management	Database Management System
Deployment	Live Web Server



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Algorithms Used

- **First-Come, First-Served (FCFS)** – Organizes charging requests according to their arrival order.
- **Greedy Approach** – Selects an appropriate available charging slot based on the current scheduling condition.
- **Priority-Based Scheduling** – Handles charging requests according to assigned priority when required.
- **Sequential Search** – Searches the available charging slots to identify a suitable slot.
- **Sorting Technique** – Arranges charging requests based on factors such as arrival time or priority.



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Future Scope and Applications

Future Scope

- AI-based prediction of charging demand during peak hours.
- Dynamic slot scheduling based on real-time availability.
- Mobile application for booking and monitoring charging slots.
- Real-time notifications for slot allocation and queue status.
- Smart energy management for efficient charging.

Applications

- EV charging stations and charging hubs.
- Shopping malls and commercial parking areas.
- Colleges and office campuses.
- Smart parking facilities with EV charging.
- Smart city EV infrastructure management.



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Conclusion

- The Smart EV Charging Station Queue and Slot Scheduling System provides an organized solution for managing EV charging services.
- It enables users to manage their profiles and vehicle information efficiently.
- Charging stations and available slots can be monitored and managed systematically.
- Queue management and automated slot allocation help reduce waiting time and improve slot utilization.
- The system provides a centralized dashboard and reports for better monitoring and decision-making.



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THANK YOU!